

MYC-C7Z010-20-V2

Hardware Design Guide



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1. Overview

This document is intended to help hardware engineers design board level circuits based on the MYC-C7Z010-20-V2 core module. Please fully understand the contents of this document before starting your design. The document contains common information such as reference design notes, Layout suggestions, and design check items to assist hardware engineers in design.

The references cited in this document are from the MYIR website and are included in the MYC-C7Z010-20-V2 product hardware collection, which can be downloaded at <http://down.myir-tech.com>

MYIR also provides resources to speed up your design:

- Core Board/Development Board Product manual;
- Development board schematic source file;
- User guides to related devices.

1.1. Supported Products

This document applies to all models of MYC-C7Z010-20-V2 series Core boards.

1.2. Disclaimer

- Some of the reference circuits in the queue documentation are based on MYIR electronic Development board and cannot be guaranteed to work in all application scenarios. If your product has special requirements on application scenarios or technical specifications, adjust the design based on actual conditions.
- The reference circuits and Layout suggestions in the lent document are for reference only and do not necessarily include all the precautions, so you can adjust them according to the actual situation.
- Buy no liability whatsoever whatsoever whatsoever for the proposals in any document.

2. Power Supply circuit design

The design of the power supply system is of vital importance in the design of embedded products, engineers need to consider not only the basic electrical parameters of power itself, but also the stability of the power supply design, such as electromagnetic compatibility, temperature range, safety design, design of three factors, such as, any neglect of factors may lead to the entire system cannot work normally. Before starting to design a power supply system for a new product, engineers should thoroughly understand the actual requirements of the entire system, and comprehensively demonstrate the feasible design scheme based on cost and efficiency, and select an appropriate power supply method for the system. If you have higher requirements for heat dissipation, you can use a radiator.

2.1. Reference circuit

Core-board needs to provide a voltage of 5V for normal operation. The average current is about 2A under full load condition, and the peak current will reach 2.5A. Considering the electricity product instantaneous power consumption is larger, and the performance of the circuit itself under the condition of high temperature will be reduced, if the power is not enough can lead to system to start, so the power supply design should have a certain margin to ensure system stable and reliable work, it is recommended to use more than 3A power supply chip separately to the Core board power supply. It is not recommended to use this power chip to drive loads outside the Core board, especially some high-power load devices.

The power supply chip can be LDO or DCDC. LDO has the advantages of simple use, low cost, small electromagnetic interference, but high calorific value. DCDC has the advantages of strong current output capacity, high conversion efficiency and low calorific value, but high electromagnetic interference. If the input voltage is close to 5V, use the LDO power chip. If the input voltage is different from 5V, use the DCDC power chip.

The 5V Core board is powered by the Baseboard. Please add the energy storage capacitor and decoupling capacitor appropriately near the 5V voltage input pin of the Core board.

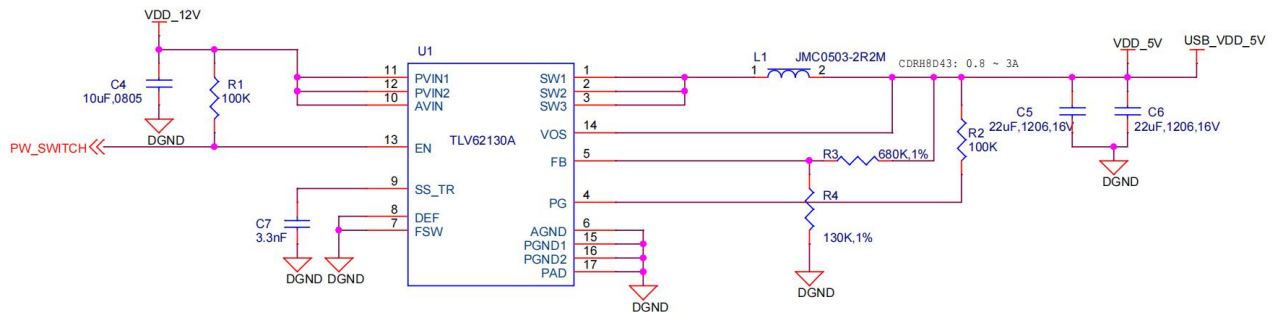


Figure 2-1 5V power supply circuit on the Core board

2.2. Power protection

To ensure the reliability of the power supply system, it is not recommended to directly supply the external untreated input voltage to the load terminals at the rear stage. Instead, a protection circuit can be designed to process the power supply before using it to improve the reliability and security of the input power supply and reduce electromagnetic interference. The Core board input power in the reference design is 5V as an example. The value of the input power should be determined according to your actual requirements.

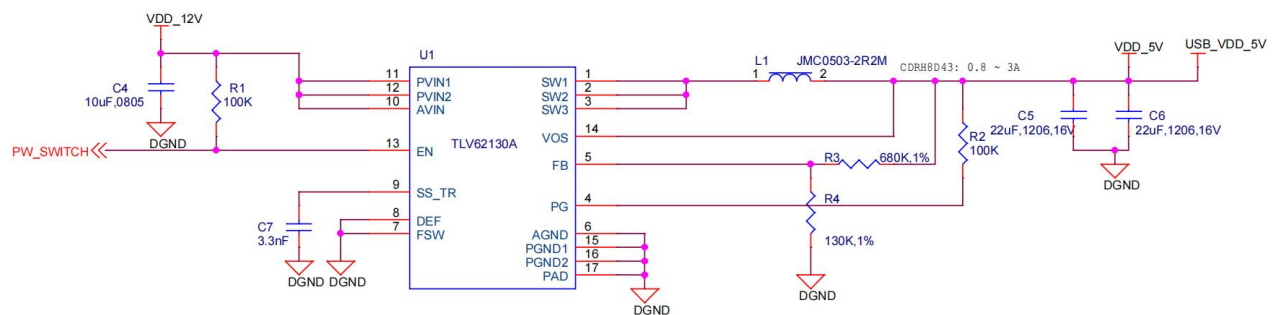


Figure 2-2 General power input circuit

DCIN<=24V

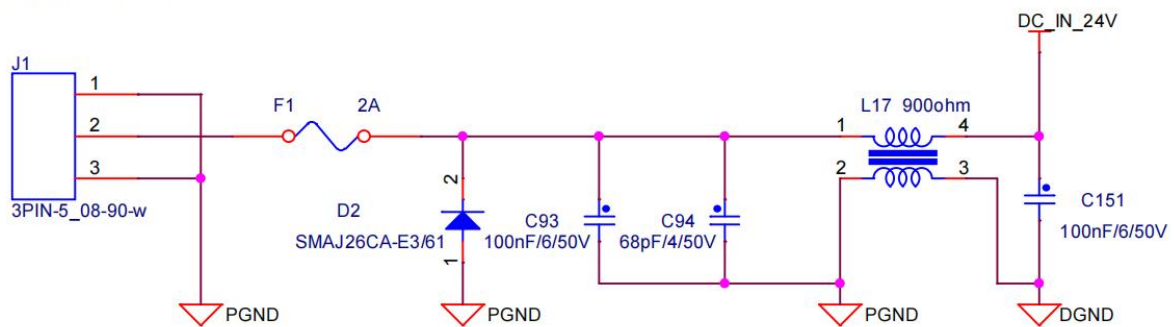


Figure 2-3 Power input circuit in noise-sensitive situations

2.3. Electricity order

The Baseboard provides 5V voltage to the Core board of MYC-C7Z010-20-V2.

Core-board drives the EN pin through Power GOOD signal to make the chip MP2143DJ work and output an appropriate voltage. The corresponding voltage can be selected as the resistance value of the feedback resistor according to the demand.

2.4. Layout recommendation

- The distance between different power planes shall be at least 20mil;
- Widen the width of power line and ground wire as far as possible to meet the required rated current value, and the width of feedback signal should not be too narrow, suggesting more than 10mil;
- If DCDC is used, the signal line is not recommended in the area below the inductance;
- If DCDC is used, the current loop path should be as short as possible, and the inductor and capacitor should be placed as close to the chip as possible, i.e. the red and green paths in the figure below.

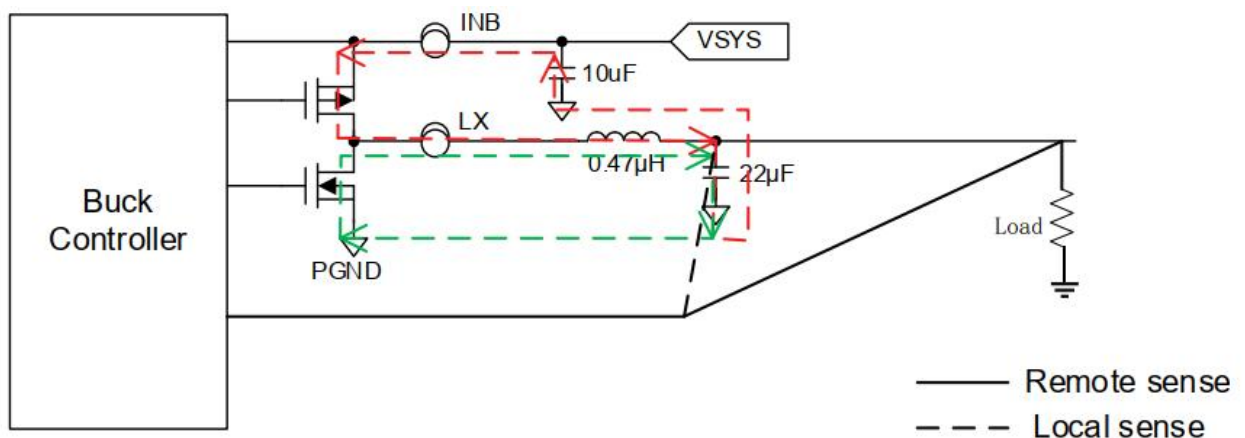


Figure 2-4 Backflow path of DCDC current

- e) If LDO is used, pay attention to the thermal resistance of LDO chip, because the thermal loss of LDO chip is relatively high, it is recommended to add grounding pads and make more grounding holes on the pads;
- f) Choose the capacitance of small ESR as far as possible
- g) The power chip with digital ground and analog ground shall be separated from each other and only connected at a single point at the input of the main power supply. The analog ground shall not be connected to the grounding pad.

3. System configuration circuit design

After the power-on reset, the MYC-C7Z010-20-V2 series processor will first execute the program in the chip's internal Boot ROM when starting up. Boot ROM startup by reading Boot pin high and low level into different Boot source.

There are two pins that determine the Boot source. The Boot Pin (1:2) is SD_BOOT_CONF_H and S_JTAG_H respectively. These pins add a pull-up or pull-down design to Core-board.

Boot Pin (1:2)	Initial Boot Source	Explain
0 0	JTAG	Boot from JTAG
0 1	QSPI	Default
1 1	Micro SD	Boot from Micro SD

Table 3-1 Boot configuration

3.1. Reference circuit

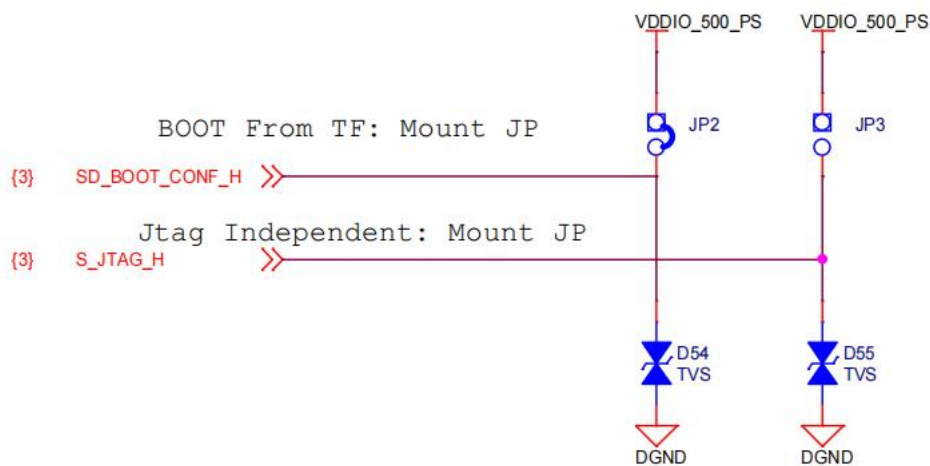


Figure 3-1 Boot configuration circuit

4. Reset and key circuit design

MYC-C7Z010-20-V2 The Core board provides special pin for reset, which is PS_WARMRESET_INn reset. It is recommended to connect it. And Baseboard can be matched with key reset. In the circuit, resistance and capacitance can be used to form a simple RC filter to filter out the jitter interference when the key is pressed, while avoiding the interference introduced from the key to affect the reset signal. In the harsh electromagnetic environment, in order to eliminate the electrostatic interference from the key to ensure more reliable operation of the system, another ESD device can be connected in parallel. If there are more stringent requirements for chattering elimination, logic circuits such as RS flip-flops can be considered to build reset circuits.

PS_MIO7 control the reset pins of USB and network PHY respectively. When the two pins are powered on, the reset pins are responsible for the above chips. If they are not in position, the above chips may not work properly. The Core board uses the SN74LVC2G07DCK to control the reset pin of the Baseboard network port and USB, which can improve the utilization rate of I/O. This method can be used when I/O is insufficient.

Special function pins	Explain
WDT_RESETN	Reset pin after power failure. RC reset circuit or hardware watchdog reset chip can be used to reset the output.
PS_WARMRESET_INn	And the Baseboard can be matched with key reset, to eliminate static electricity can be connected to an ESD diode.

Table 4-1 Description of reset signals

4.1. Reference circuit

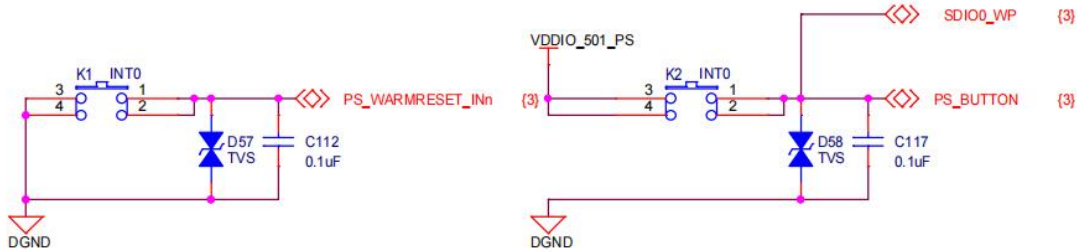


Figure 4-1 Resetting the reference circuit

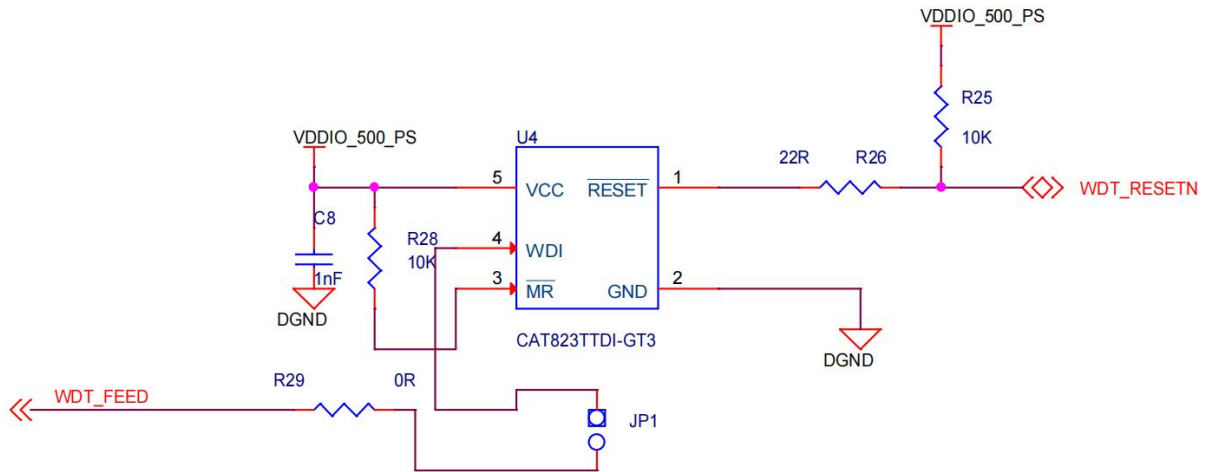


Figure 4-2 Resetting the reference circuit using an external hardware watchdog

4.2. Layout suggest

- Reset signal line width should not be too narrow, it is recommended not less than 8mil;
- Reset signal is a sensitive signal, and it is recommended to deal with it unconditionally;
- Place TVS as close to the button as possible.

5. Interface circuit design

5.1. SDMMC

MYC-C7Z010-20-V2 Core board support leads to the MMC interface with SDIO0, SDIO1 with EMMC. TXS02612 chip can be used for expansion, a single SDIO port and two SDIO peripherals interface. It is recommended to use MMC1 interface for Micro SD circuits and MMC2 to use peripherals with SDIO WIFI or other SDIO interfaces.

When designing SD/SDIO/MMC card interface circuit, it is necessary to pass through TXS02612 expander with level conversion and then connect these interfaces to SD/MMC card accordingly. Figure 5-2 shows the MMC1 circuit.

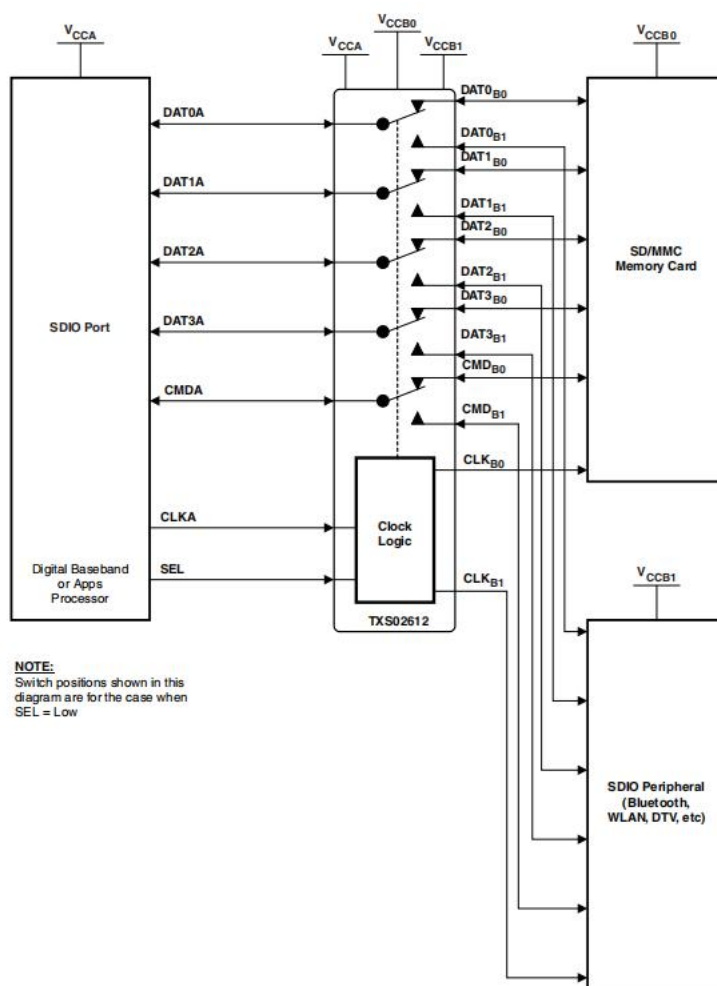


Figure 5-1 Schematic diagram of expanding ports

5.1.1. Reference circuit

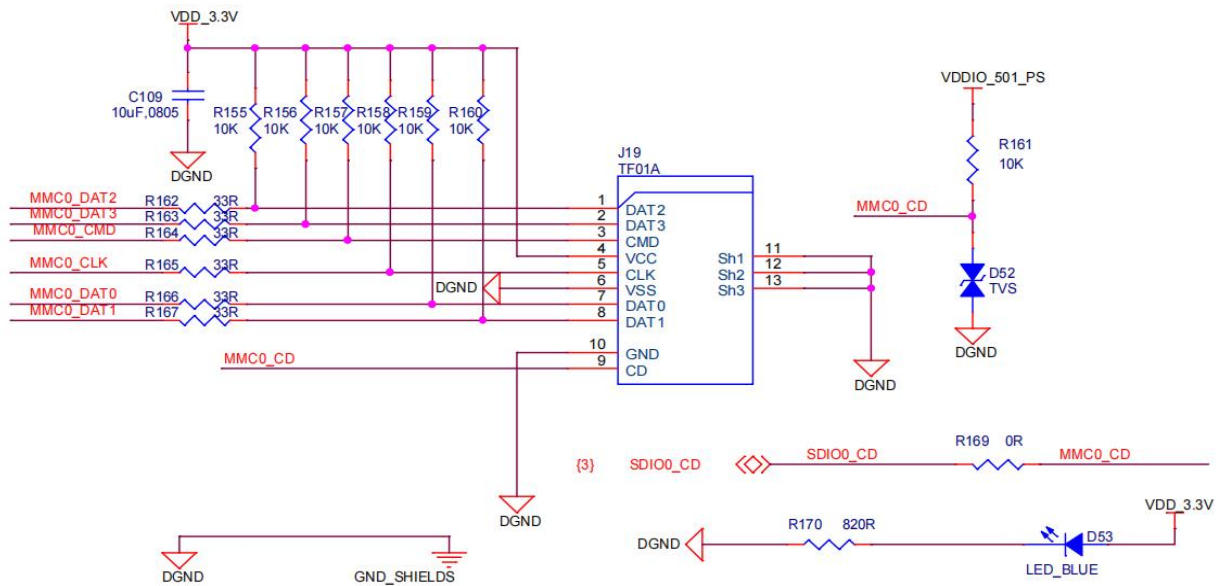


Figure 5-2 Reference circuit of the SD card

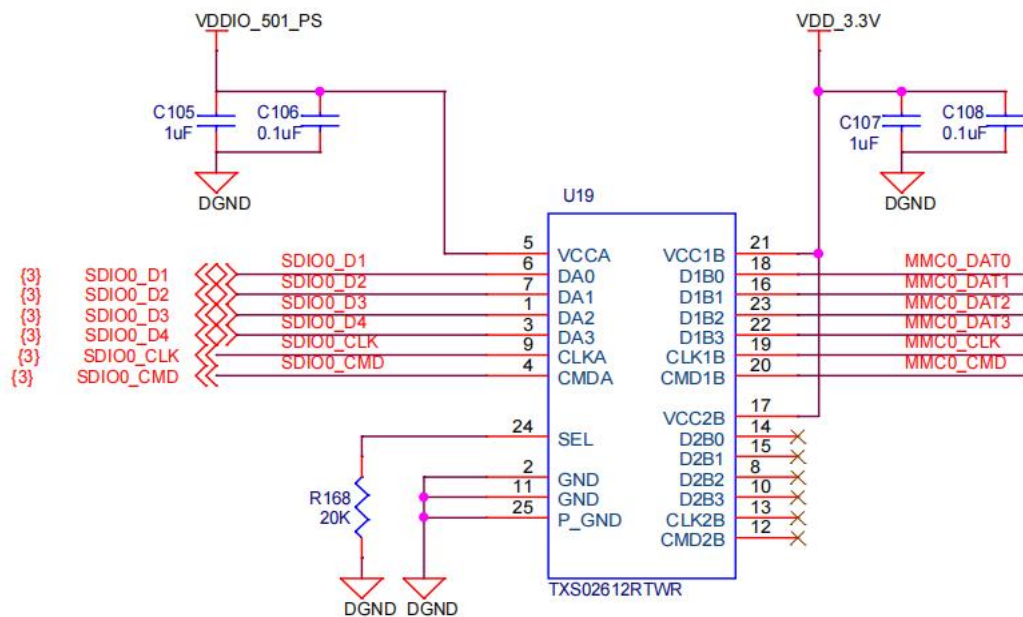


Figure 5-3 Reference circuit of the expansion port

5.1.2. Layout suggest

- SDMMC traces keep 50Ω impedance. clock signals are recommended to be surrounded with ground.
- Trace match the data line and control line to be within 100 mils;
- Clock signals are recommended to be surrounded with ground. If not, ensure the distance between the clock signal and other signals follow 3W rule.

5.2. UART

The Core board of MYC-C7Z010-20-V2 supports up to eight asynchronous serial ports. Due to the pin reuse of the chip, the Core board only uses 3 channels of serial port by default, among which UART1 has flow control (RTS and CTS signal) function, and the other 2 channels of UART only have TXD and RXD by default.

In the reference design, UART signal has the following usages: UART to RS232, UART to RS485, UART signal directly connected for debugging. considering the harsh environment of the industrial site, UART to RS232 and UART to RS485 circuit are used in isolation connection.

In the debugging serial port of UART direct continuous operation, the UART signal itself is TTL3.3V level, so UART to USB converter is needed to connect with PC conveniently. The converter accessories can be used by MYIR Company.



Figure 5-4 UART conversion cable

5.2.1. UART to RS232

In the reference circuit, THE UART to RS232 conversion chip is SP3232EY-L of EXAR Company, and TXS0102DCTR chip used for signal level conversion chip . After conversion, RS232_TX/RX can be directly connected to the DSUB-9 connector. and then with DSUB-9 serial line for serial output printing.

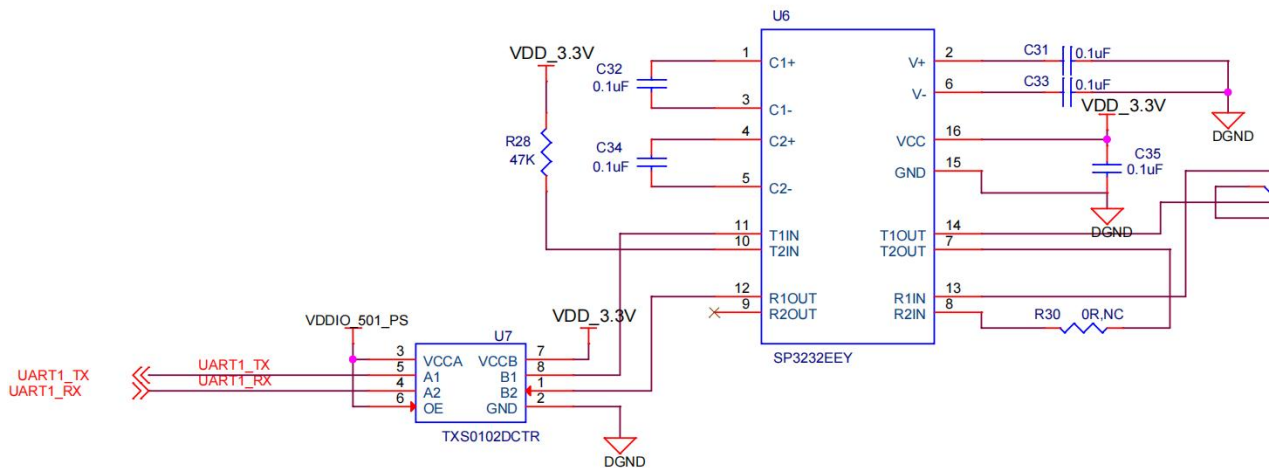


Figure 5-5 RS232 reference circuit

5.2.2. Layout suggest

- Keep sufficient spacing between signal and power plane before and after isolation;
- Differential signal isometric control, difference internal error $\pm 30\text{mil}$, spacing between adjacent difference pairs is more than 3W;

5.3. USB

MYC-C7Z010-20-V2 The Core board supports two USB2.0 HOST interfaces, or one USB2.0 HOST interface and one USB2.0 OTG interface. If user want to use the OTG function of USB, they can use Micro USB port or Type C USB port. It is recommended that a fuse be added between the power supply of the USB module and VDD_5V. TVS tube and common-mode inductor are recommended for USB signal.

5.3.1. Reference circuit

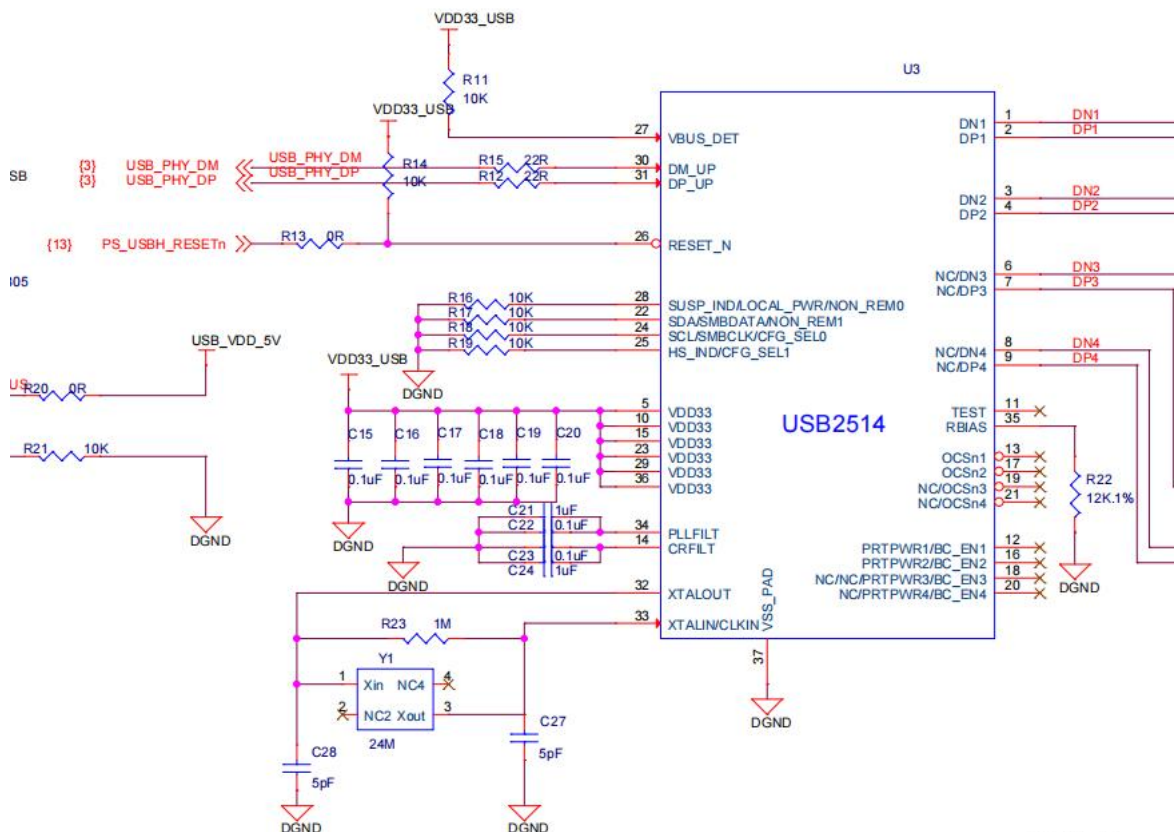


Figure 5-6 Interface reference circuit

5.3.2. Layout recommendation

- USB signal wiring for isometric control, error range $\pm 25\text{mil}$;
- Distance between adjacent differential pairs of USB signal wiring is more than 3W.

- c) USB signal cable should be as short as possible;
- d) USB signal should not change layers as far as possible. If it changes layers, the GND backflow hole should be placed within 50mil from the change layer hole;
- e) Ensure that the reference plane is continuous and USB signal is not divided across;
- f) USB signal wiring is recommended at the TOP/BOTTOM layer;
- g) KEEP USB signals away from other clocks and digital signals.

5.4. Ethernet

MYC-C7Z010-20-V2 Core board supports two Gigabit network ports. User need to design PHY chip circuit and network port socket to achieve network communication. Gigabit Ethernet Controller (GEM) implements 10/100/1000 Mb/s Ethernet mac, is compliant with IEEE 802.3-2008 standards, and is capable of running in half-duplex or full-duplex mode at all three speeds. PS comes with two Gigabit Ethernet controllers. Each controller can be configured independently. PS ETH0 connects to YT8531, and ETH1 connects to USB0.

4.1.1. UTP (UTP<->RGMII / UTP<->SGMII) Application

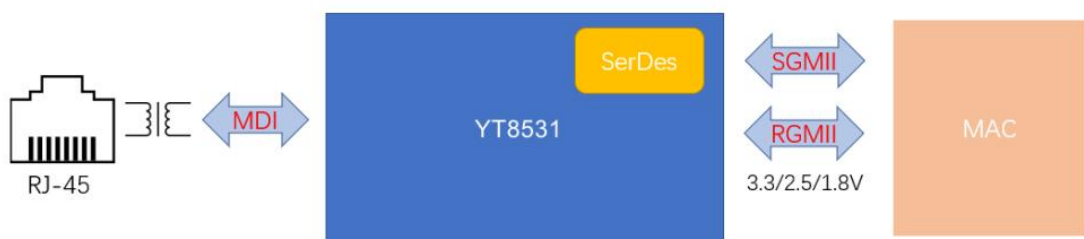


Figure 3. UTP (UTP<->RGMII / UTP<->SGMII) Application

Figure 5-7 Ethernet link

5.4.1. Reference design

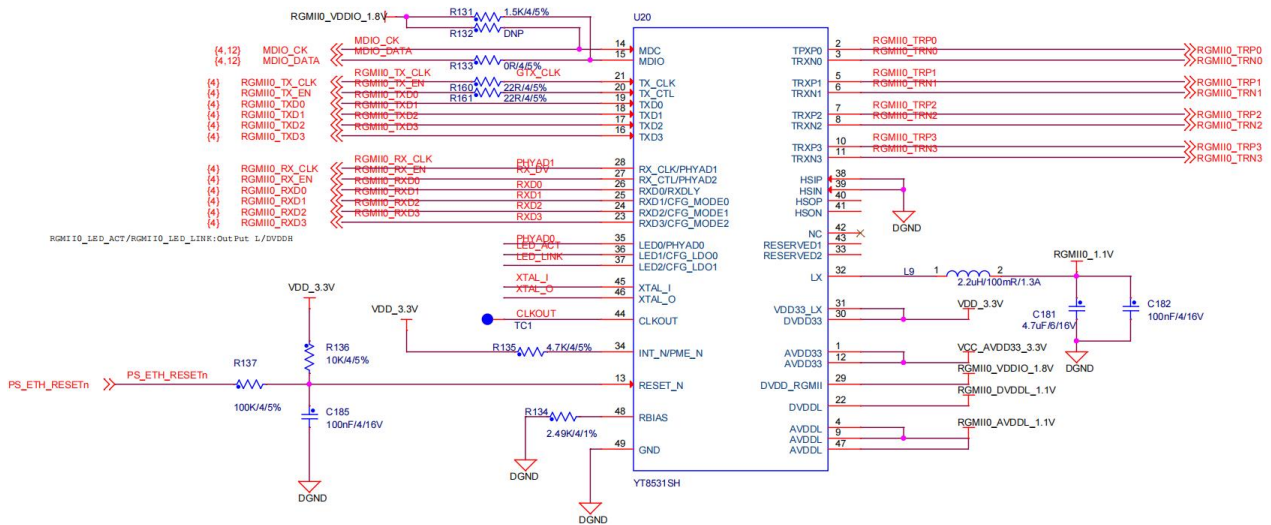


Figure 5-8 Ethernet reference circuit

5.4.2. Layout suggest

- PHY chip is placed close to the core plate and away from the network transformer;
- The network transformer is placed close to the RJ45 interface;
- The power pin decoupling capacitor of the PHY chip is placed close to the PHY chip;
- It is recommended that the pin of the network port be directly connected to the ground of the chassis;
- RGMII signal isometric control, error $\pm 25\text{mil}$, line spacing more than $2W$;
- Network differential signal isometric control, difference internal error $\pm 30\text{mil}$, spacing of adjacent difference pairs more than $3W$;

5.5. CAN

The MYC-C7Z010-20-V2 supports two CAN controllers. Comply with ISO 11898-1, CAN 2.0A, CAN 2.0B standards. Supports standard frames (11-bit identification) and extended frames (29-bit identification). Supports 1mb /s bit rate. If users need to use CAN interface to develop products, please contact sales for customized core boards.

CAN controller CAN communicate with CAN transceiver by designing related circuit. Can-bus communication reference circuit is shown in the figure below, which adopts high-speed CAN transceiver module TJA1050T. The module can be up to 1Mbaud, very low electromagnetic emission (EME). It is suitable for wide common-mode differential receiver with high EMI and can be used with ADUM1201 dual channel digital isolator to ensure the reliability of communication.

In order to achieve complete electrical isolation, it is also necessary to use isolation power module to generate a special isolation power supply for CAN. B0505s-1wr2 isolation power supply of Jin Sheng yang is used in the reference circuit to generate 5V isolation voltage for the CAN transceiver circuit. Notice the psu has a minimum load value and a 10K resistance is required as the default load. For details about related circuits, see section 5.2.

CANH and CANL signals need to be well protected. TVS tube is added in the normal environment. The model of TVS tube in the reference circuit is SMBJ7.0CA-E3/52. Protective devices such as common mode inductors, gas discharge tubes, and surge suppressors should also be considered if the bus is used in a harsh environment.

5.5.1. Reference circuit

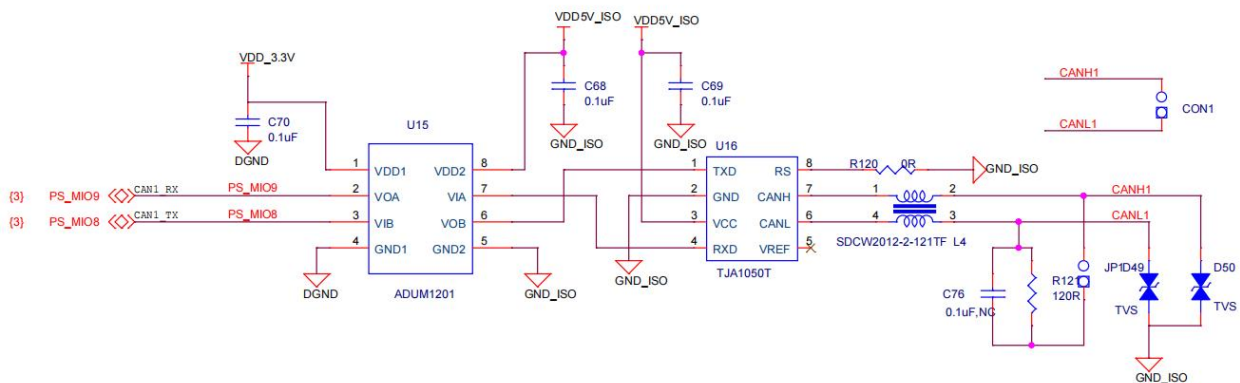


Figure 5-9 Isolating the CAN sending and receiving reference circuit

5.5.2. Layout suggest

- R12 (120Ω) \C76 placed close to the transceiver, D50 placed close to the CAN connector;
- CAN signal should be as long as possible with an error of $\pm 300\text{mil}$;
- Isolate the area under the power supply, including the inner layer, and do not wire it.

5.6. JTAG

The standard JTAG interface consists of four lines :TMS, TCK, TDI and TDO, which are mode selection, clock, data input and data output lines respectively. PL-jtag Interface clock PL-JTAG uses the JTAG port to connect to the XADC interface and uses the JTAG clock TCK. JTAG boot mode is considered a slave boot mode and is always an insecure boot mode. The JTAG chain can be configured in cascading and independent modes.

In standalone JTAG mode, users can only access Xilinx TAP through PLTDO/TMS/TCK/TDI. When users access ARM DAP, they must use PJTAG signal, as shown in the figure below. There are two options for routing PJTAG signals to the chip pinout: EMIO to PL Select IO or MIO.

Table 27-1: PJTAG Signals

Signal Name	via EMIO		via MIO	
	Signal	I/O	Pin	I/O
TCK	EMIOPJTAGTCK	I	MIO 12, 24, 36 or 48	I
TMS	EMIOPJTAGTMS	I	MIO 13, 25, 37 or 49	I
TDI	EMIOPJTAGTDI	I	MIO 10, 22, 34 or 46	I
TDO	EMIOPJTAGTDO	O	MIO 11, 23, 35 or 47	O
TDO 3-state	EMIOPJTAGTDTN	O		

5.6.1. Reference circuit

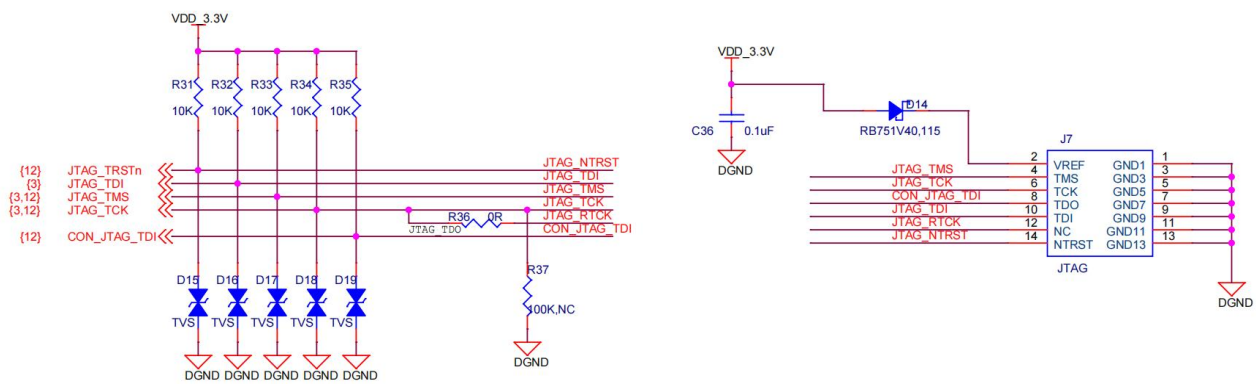


Figure 5-10 JTAG reference circuit

5.6.2. Layout suggest

- TVS diode should be added to the signal for electrostatic protection.
- The clock and data line shall follow the 3W principle;
- To ensure consistency and continuity of the reference layer of the signal.

5.7. SPI

MYC-C7Z010-20-V2 Core board supports a maximum of two SPI controllers, supporting master/slave mode. SPI signals include SPI_CLK, SPI_SS, SPI_MOSI and SPI_MISO. During design, the relationship between master and slave devices should be confirmed first, and then the direction of MOSI and MISO signals should be confirmed.

The Quad-SPI Flash controller is part of the input/output peripherals (IOP) located inside PS. It is used to access multi-bit serial Flash storage devices for high throughput and low pin count applications.

The controller operates in one of three modes: I/O mode, linear addressing mode, and legacy SPI mode. In I/O mode, the software interacts closely with the Flash device protocol. The software uses four TXD registers to write flash commands and data to the controller. The software reads the RXD registers that contain the data received from the flash device.

Only one QSPI interface is configured in the design of the core board. To use more QSPI interfaces, consult the chip manual. The processor is directly connected to the Flash, and the corresponding I/O port should be connected to a matching pull-up or pull-down resistor to determine the corresponding level.

5.7.1. Reference frame

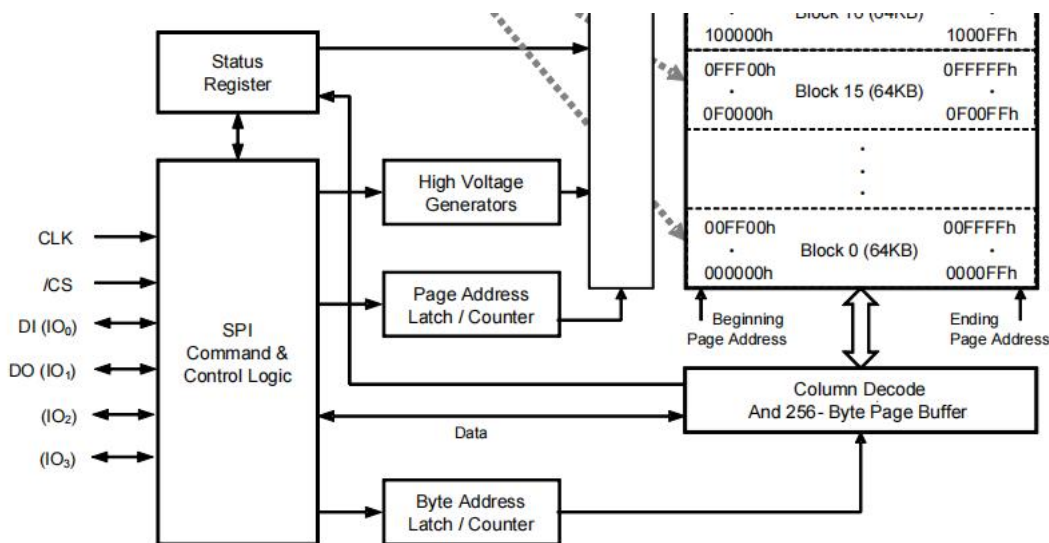


Figure 5-11 SPI reference frame

5.7.2. Layout suggest

- SPI signal single-end impedance control is 50Ω;
- SPI clock and data line follow the 3W principle;
- Ensure that the reference layer of the signal is continuous.

5.8. Generic I/O interface

The Universal I/O (GPIO) peripheral provides software to observe and control 54 device pins through the MIO module. It also provides programmable logic access to 64 inputs (PL) and 128 outputs to PL through the EMIO interface. GPIO is organized into four sets of registers that group related interface signals.

Each GPIO is independently and dynamically programmed as input, output, or interrupt sensing. Software can read all GPIO values within Banks using a single load instruction, or write data to one or more GPIO(within the scope of the GPIO) using a single storage instruction. GPIO control registers and state registers are memory-mapped at base address 0xE000_A000.

Features: Device pin routing for up to 54 GPIO signals through MIO output is 3 state, 192 GPIO signals between PS and PL through EMIO, 64 input, 128 output (64 true output and 64 output enabled).

Each GPIO feature can be dynamically programmed on an individual or group basis, enabling, Bit or Bank data writing, output enabling and direction control.

State reading of programmable interrupts, raw interrupts and masked interrupts on a separate GPIO basis; Positive side, negative side, arbitrary side, high level, low level sensitivity.

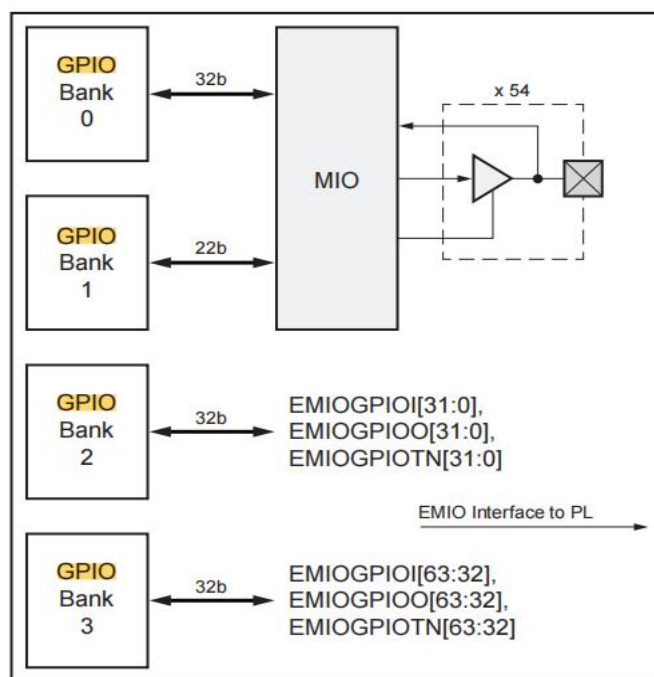


Figure 5-12 GPIO block diagram

5.8.1. Reference circuit

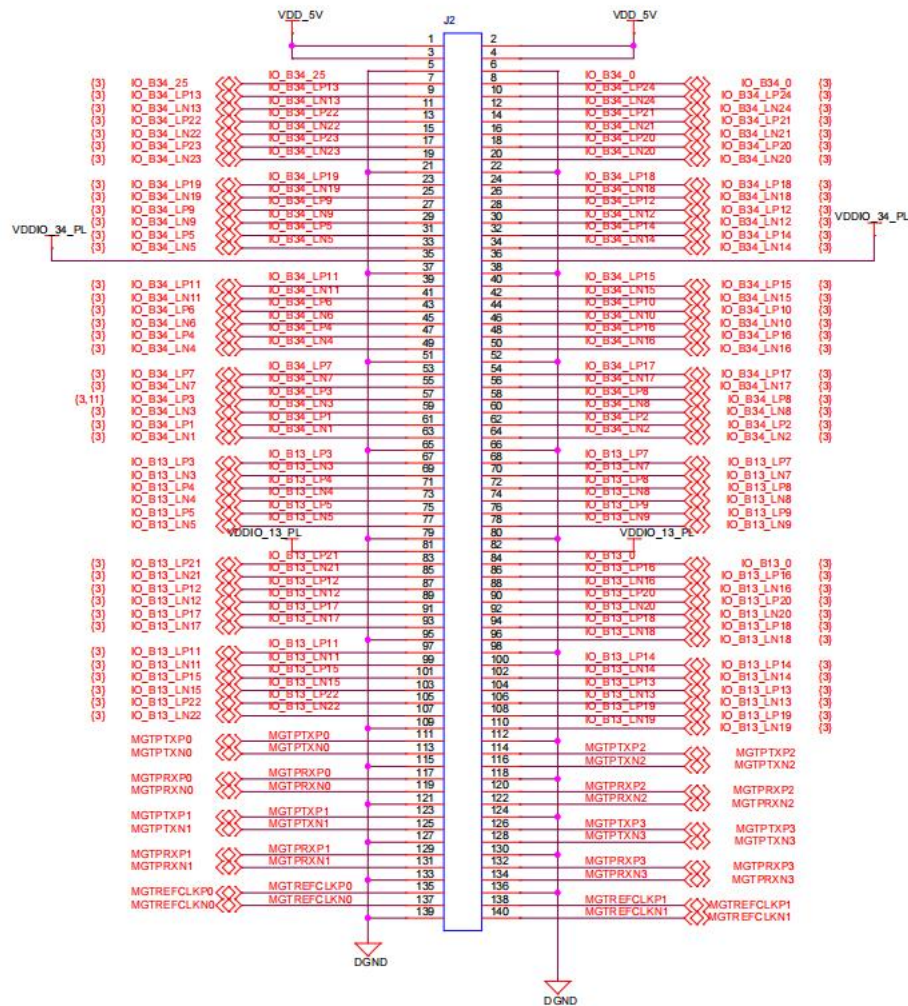


Figure 5-13 Reference circuit of the GPIO port

5.8.2. Layout suggest

- The capacitance resistor of the GPIO port is placed close to each other.
- The signal wiring of GPIO port is controlled with equal length, and the error range is $\pm 100\text{mil}$; (Note: The Package Pin delay was not added into THE PCB constraint during PCB design of core board, and the actual wiring length is not equal, so the customer needs to add Pin delay)

5.9. PS MIO I/Os Port

PS MIOPS MIO I/O buffers are divided into two voltage domains. Each MIO is independently programmable in each domain. Typically, each pin is assigned a function.

Two I/O voltage groups: Bank 0 voltage group is composed of pin 0:15; The Bank 1 voltage group consists of pin 16:53.

MIO voltage levels can be programmed by Bank, 1.8 and 2.5/3.3 volts; CMOS single-ended or HSTL differential receiving mode.

Interface frequency: The clock frequency of the interface usually depends on the speed level of the device and whether the interface is routed through MIO or EMIO. Table 5-19 lists the possible routes of the interfaces. The maximum clock frequency that can be used for each speed level and routing path is defined in the ZynQ-7000 AP SoC data table.

Table 2-4: MIO-at-a-Glance

[illegible]

Figure 5-14 I/O ports

5.9.1. Reference circuit

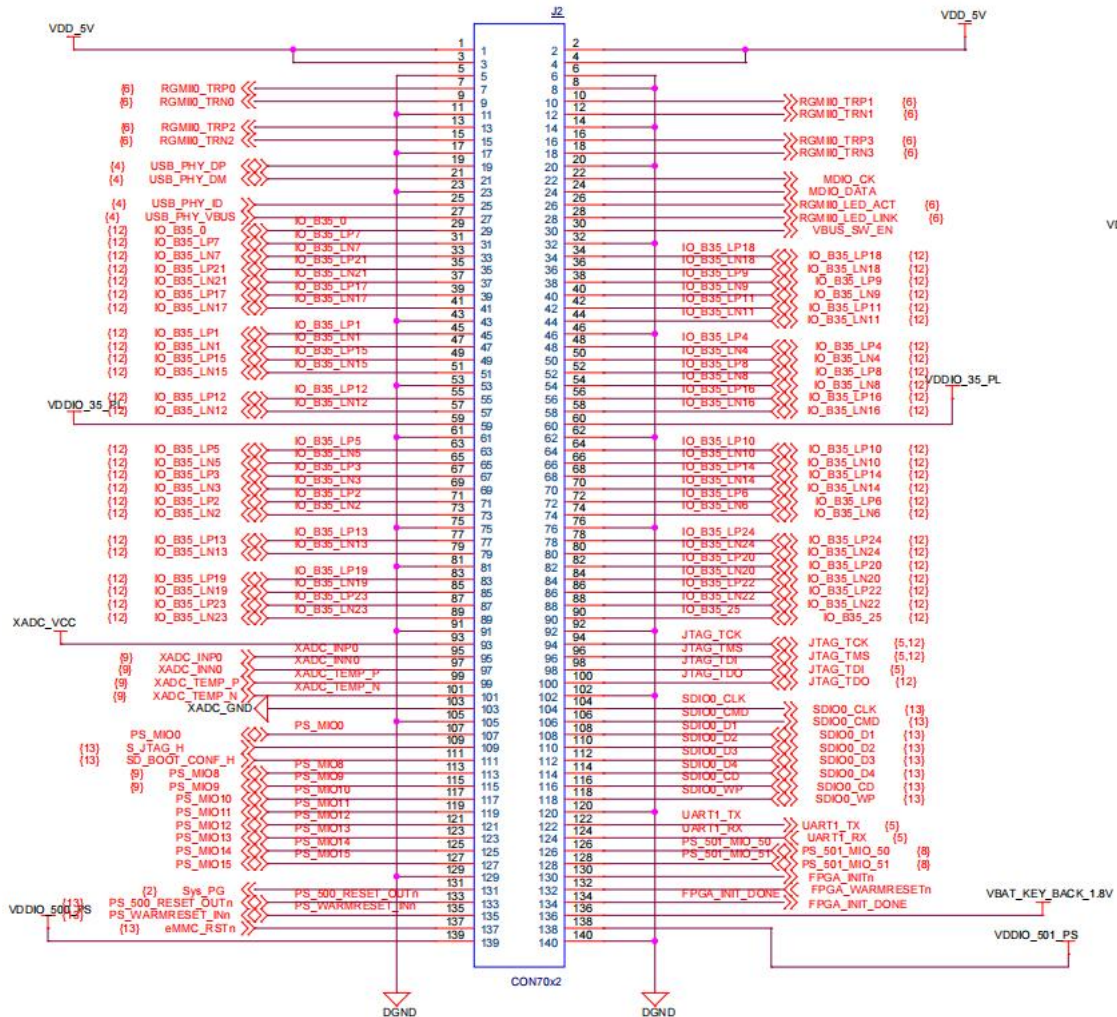


Figure 5-15 Interface reference circuit

5.9.2. Layout suggest

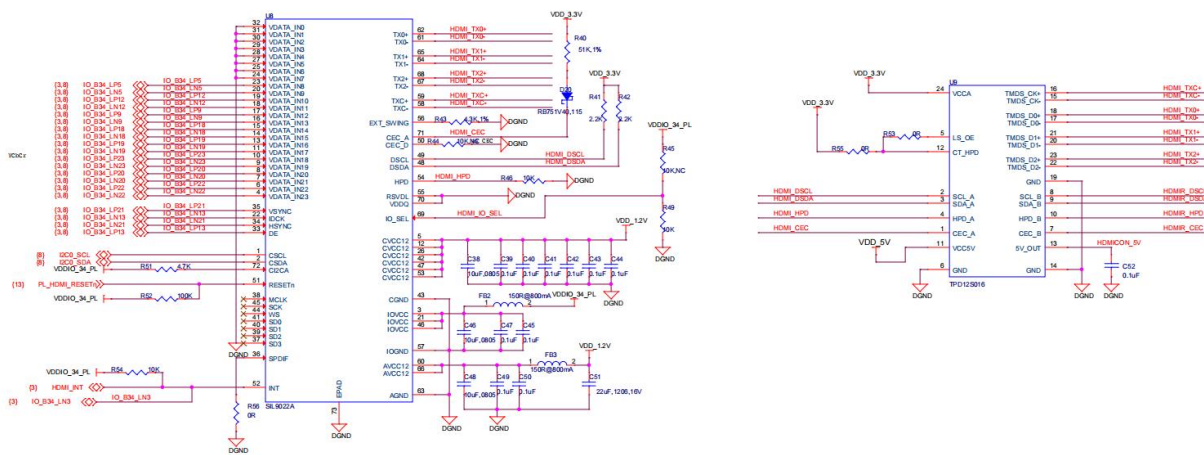
- The resistor capacitor is placed close to the IO;
- The signal wiring of IO port is controlled with equal length, and the error range is $\pm 100\text{mil}$;

5.10. HDMI

MYC-C7Z010-20-V2 is designed with 1 channel HDMI port, which is controlled by GPIO of BANK. Universal I/O (GPIO) peripherals provide observation and control software for up to 54 device pins through the MIO module. It also provides access to 64 inputs to programmable logic (PL) and 128 outputs to PL through the EMIO interface. GPIO is organized into four sets of registers that group related interface signals.

The design uses an HDMI transceiver: SIL9022A, which is a pre-programmed HDCP key and has fully self-sorting HDCP detection and authentication, including SHA-1 for Repeaters. Devices support high bandwidth digital Content Protection (HDCP) for devices that require secure content delivery. Use I2C protocol for communication. It can be matched with TPD12S016 chip to realize level transfer buffer, 12-channel ESD, and can also be used as current limiting load switch.

5.10.1. Reference circuit



- d) The HDMI signal should not change the layer as far as possible. If the layer is changed, the GND backflow hole should be placed within 50mil from the layer change hole;
- e) HDMI ground pins are recommended to be directly connected to the ground of the chassis;

5.11. LCD

MYC-C7Z010-20-V2 Core board is designed with 1 channel LCD controller with 24Bit RGB signal, which supports color TFT display screen and other LCD displays. The default resolution is 800x480 (7 inch screen). You can choose my-My-TFT070CV2 LIQUID crystal display module of MYIR Electronics for use.

MY-TFT070CV2 is a 7-inch LCD capacitive touch screen. Please visit <http://www.myir-tech.com/product> for module details.

HDMI and LCD do not share RGB signals, which may cause mutual interference. It is recommended to use them separately.



Figure 5-20 MY-TFT070CV2 LCD display module

NO	Item	Parameter	Description
1	LCD size	7 Inch(16:9)	Diagonal
2	Resolution	800(H)×3(RGB)×480(v) mm	
3	Display area	95(H)× 53.86(V) mm	
4	Outline Dimension	165(w)×104(H)×5.5(T) mm	
5	Module size	177 × 106 mm	
6	Display mode	Normally White	
7	Dot pitch	0.0642(W)× 0.1790(H)mm	
8	Surface treatment	Anti-Glare	
9	Operation Temperature	-20°C – 70°C	

10	Storage Temperature	-30°C – 80°C	
11	Interface	50pin FPC	0.5mm pitch
		50pin IDC connector (25pinX2)	1.27mm pitch
12	Weight	150g	
13	Backlight power consumption	1.674W	
14	Panel power consumption	0.226W	

Figure 5-17 MY-TFT070CV2 LCD module

5.11.1. Reference circuit

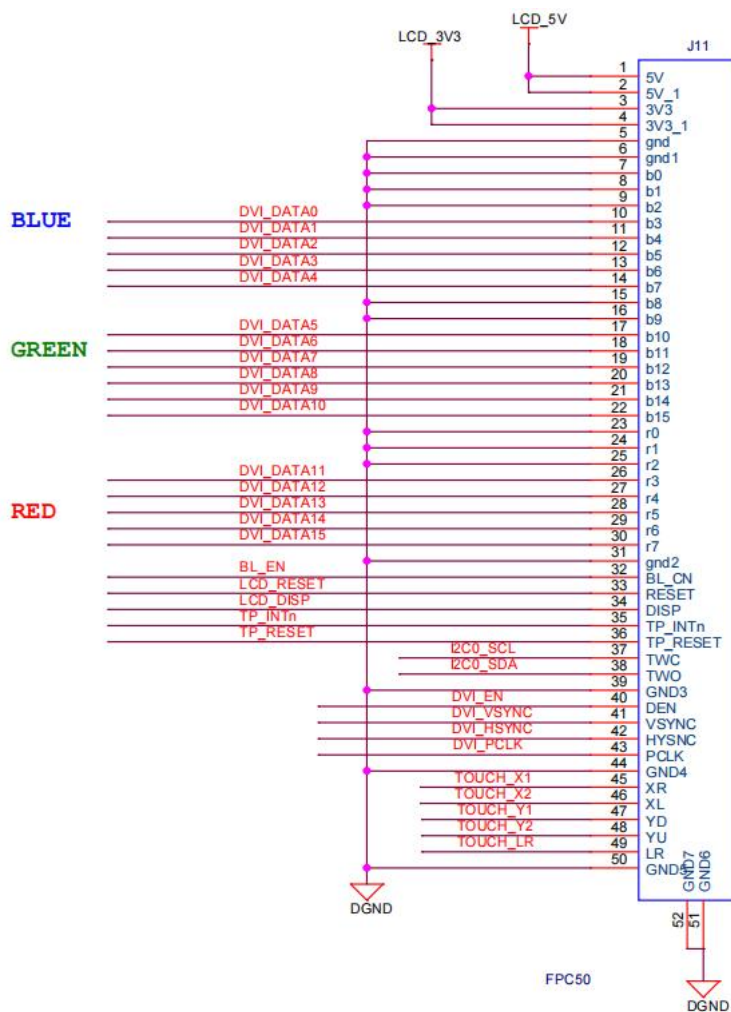


Figure 5-18 MY-TFT070CV2 LCD module

5.11.2. Layout suggest

- The 22Ω resistor is placed close to the connector of the core plate;
- LCD signal wiring for equal length control, error range $\pm 100\text{mil}$; The distance between LCD signal cables must be at least 2W.

5.12. FMC

The MYC-C7Z010/20-V2 Core board is designed with 1 channel FMC interface circuit. The FMC standard is designed to provide standard sandwich board (sub card) size, connector and module interface for FPGA on the Baseboard (carrier card). In this way, THE I/O interface is separated from THE FPGA, simplifying the design of the I/O interface module and maximizing the repeated utilization of the load card. Support up to 10Gb/s signal transmission rate, the potential total bandwidth between the sandwich board and the carrier up to 40Gb/s. FMC signal matching impedance is designed according to the requirements of 50 Ω for single end and 100 Ω for difference. The model is ASP-134603-01 of Sametc.

5.12.1. Reference circuit

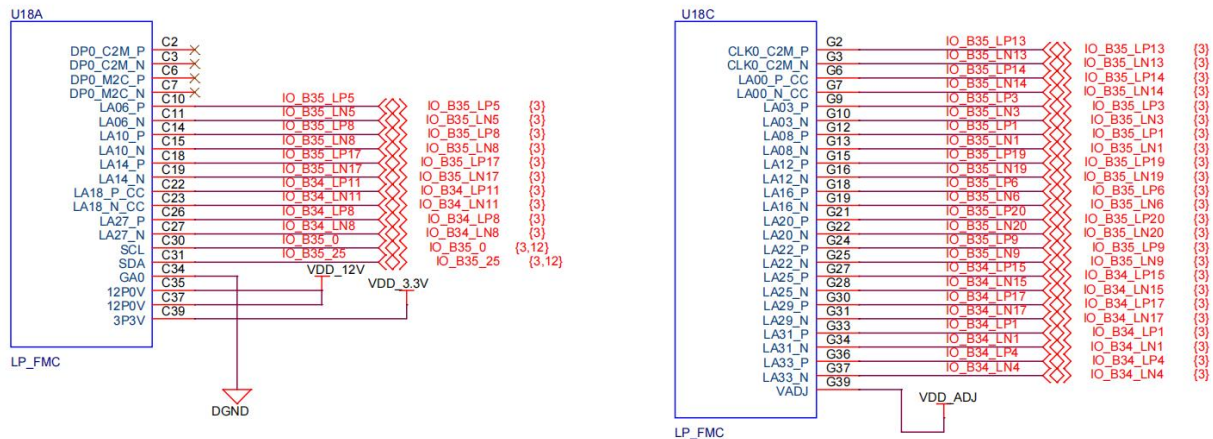


Figure 5-20 FMC reference circuit

5.12.2. Layout suggest

- Spacing between adjacent difference pairs of signal routing is more than 3W.
- The signal line should be as short as possible; Signal routing is equal length control, error range $\pm 25\text{mil}$.
- The signal should not change layers as far as possible. If it changes layers, the GND backflow hole should be placed within 50mil from the change layer hole.

5.13. RTC battery interface

The VBAT_KEY_BACK_1.8V pin (F11) of the MYC-C7Z010-20-V2 Core board is used to connect the pin of the button lithium battery. Figure 5-24 shows the power supply design for the RTC.

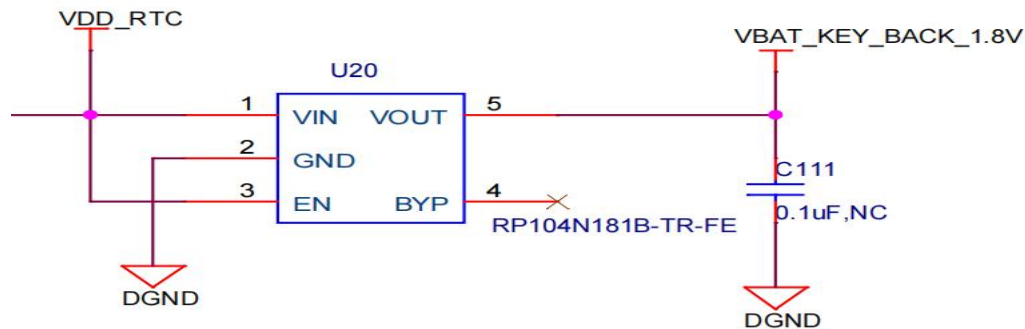


Figure 5-24 RTC power supply circuit inside the core board

J20 is a battery holder that can hold 2032 button batteries rated at 3.3V. considering the small capacity of the button battery, it is recommended to add a dedicated RTC chip to the baseboard if the user has requirements on the battery life. In addition, the button battery only supplies power to the dedicated RTC chip.

5.13.1. Reference circuit



Figure 5-25 RTC design circuit

5.13.2. Layout suggest

a) RTC standby lithium battery PCB power wiring recommended 10mil or above.

6. Design Inspection Items

6.1. Power supply design check items

Check item	Proposed projects
Core board power supply voltage	The recommended value is 5V, with an absolute value of 4.9V-5.5V
Core board power supply decoupling capacitor	47uF and above
IO level of Baseboard peripherals	The IO level of the peripheral should match the corresponding interface level of the Core board
Core board power timing	It is recommended that the Core board power supply start before the peripheral power supply
Temperature rise of power supply chip	Confirm the thermal resistance of the power supply chip, and calculate the maximum temperature rise of the power supply chip based on the power consumption of the Core board, to ensure that the final temperature is within the specified range of the power supply chip

Table 6-1 Power supply design checklist

6.2. Check the system startup

Check item	Proposed projects
Boot[2:1]Configuration of pins	The configuration depends on the startup mode of the product
PS_PWRON_RESETn	Use RC reset circuit or add external hardware watchdog reset.
Micro SD	Micro SD card for easy batch firing procedures. Recommend leaving

Table 6-2 System startup checklist

6.3. Some peripherals circuit design check matters

Classify	Check item	Proposed projects
USB	Capacitance of USB D+/D- signal ESD device	It is recommended that the capacitance of the ESD device be smaller than 2pF
	Whether the capacitor of the power supply pin is series resistance	The interface 5V capacitor requires a 1 ohm resistor in series to limit the voltage surge at the USB port
Ethernet (RGMII)	PHY chip layout	Get as close to the Core board layout as possible. Keep RGMII wiring as short as possible. RGMII sends and receives signals in separate groups, and Layout Layout within the same length +-25mil. There is no requirement between groups.
	PHY chip power supply	The PHY chip power supply is isolated by magnetic beads
	Clock signal source of the PHY chip	Use external active or passive crystal oscillator.
	Connection method of center tap on PHY side of network transformer	Depending on the type of PHY chip, it can be found in the chip manual. If the PHY is current-driven, the tap needs to be pulled up to the PHY supply voltage. If the PHY is voltage-driven, the tap does not need to be pulled up. If not found in the manual, use a reference circuit or reserved pull-up resistor
MMC	Whether DATA and CMD signals are pulled up	Requires pull-up, 47K or 10K pull-up to 3.3V
CAN	Whether the CAN circuit needs to be isolated	Scenario Complex electrical environment high reliability requirements CAN interface cable length is long. If any of the preceding conditions are met, isolate the CAN converter and its power supply circuit
UART	UART signal connection	UART signal can not be directly connected to RS232\RS485 interface, the application of special conversion chip conversion can be connected to the corresponding interface

Table 6-3 Peripherals circuit checklist

7. Explanation of common problems

7.1. Board to board connector core plate seismic poor, why not use other connection methods

If the stamp hole scheme is adopted, the advantage is that the signal connection is better, but there are also the following disadvantages:

- 1) It is difficult to disassemble;
- 2) High cost, half hole process;
- 3) There may be a virtual welding problem when welding again.

If the scheme of gold finger is adopted, it will be more convenient to plug and remove, but it also has the following disadvantages:

- 1) The Baseboard needs a high-quality base, which increases the cost;
- 2) High production cost of Goldfinger;
- 3) It cannot be used for thin and light products.

Weighing the advantages and disadvantages, MYC-C7Z010-20 Core board adopts the connection mode of stamp hole, which does not have the above disadvantages.

7.2. Treatment method for short circuit of module pin after patch

- 1) Check whether the solder paste on the solder pad of the module is connected with tin;
- 2) Check whether the patch of the module is offset;
- 3) If it is found that it has been pasted crooked before the furnace inspection, it cannot be adjusted and straightened manually. The tin paste at the bottom will move the tin, so it is suggested to remove and paste it again.
- 4) Check whether the SMT process and PCB design meet the requirements of Chapter 8 in MYC-C7Z010-20 Product Manual.

Appendix A

Warranty & Technical Support Services

MYIR Electronics Limited is a global provider of ARM hardware and software tools, design solutions for embedded applications. We support our customers in a wide range of services to accelerate your time to market.

MYIR is an ARM Connected Community Member and work closely with ARM and many semiconductor vendors. We sell products ranging from board level products such as development boards, single board computers and CPU modules to help with your evaluation, prototype, and system integration or creating your own applications. Our products are used widely in industrial control, medical devices, consumer electronic, telecommunication systems, Human Machine Interface (HMI) and more other embedded applications. MYIR has an experienced team and provides custom design services based on ARM processors to help customers make your idea a reality.

The contents below introduce to customers the warranty and technical support services provided by MYIR as well as the matters needing attention in using MYIR' s products.

Service Guarantee

MYIR regards the product quality as the life of an enterprise. We strictly check and control the Core board design, the procurement of components, production control, product testing, packaging, shipping and other aspects and strive to provide products with best quality to customers. We believe that only quality products and excellent services can ensure the long-term cooperation and mutual benefit.

Price

MYIR insists on providing customers with the most valuable products. We do not pursue excess profits which we think only for short-time cooperation. Instead, we hope to establish long-term cooperation and win-win business with customers. So we will offer reasonable prices in the hope of making the business greater with the customers together hand in hand.

Delivery Time

MYIR will always keep a certain stock for its regular products. If your order quantity is less than the amount of inventory, the delivery time would be within three days; if your order quantity is greater than the number of inventory, the delivery time would be always four to six weeks. If for any urgent delivery, we can negotiate with customer and try to supply the goods in advance.

Technical Support

MYIR has a professional technical support team. Customer can contact us by email (support@myirtech.com), we will try to reply you within 48 hours. For mass production and customized products, we will specify person to follow the case and ensure the smooth production.

After-sale Service

MYIR offers one year free technical support and after-sales maintenance service from the purchase date. The service covers:

Technical support service

MYIR offers technical support for the hardware and software materials which have provided to customers;

- To help customers compile and run the source code we offer;
- To help customers solve problems occurred during operations if users follow the user manual documents;
- To judge whether the failure exists;
- To provide free software upgrading service.
- However, the following situations are not included in the scope of our free technical support service:
 - Hardware or software problems occurred during customers' own development;
 - Problems occurred when customers compile or run the OS which is tailored by themselves;
 - Problems occurred during customers' own applications development;

- Problems occurred during the modification of MYIR' s software source code.

After-sales maintenance service

The products except LCD, which are not used properly, will take the twelve months free maintenance service since the purchase date. But following situations are not included in the scope of our free maintenance service:

- The warranty period is expired;
- The customer cannot provide proof-of-purchase or the product has no serial number;
- The customer has not followed the instruction of the manual which has caused the damage the product;
- Due to the natural disasters (unexpected matters), or natural attrition of the components, or unexpected matters leads the defects of appearance/function;
- Due to the power supply, bump, leaking of the roof, pets, moist, impurities into the boards, all those reasons which have caused the damage of the products or defects of appearance;
- Due to unauthorized weld or dismantle parts or repair the products which has caused the damage of the products or defects of appearance;
- Due to unauthorized installation of the software, system or incorrect configuration or computer virus which has caused the damage of products.

Warm tips:

- MYIR does not supply maintenance service to LCD. We suggest the customer first check the LCD when receiving the goods. In case the LCD cannot run or no display, customer should contact MYIR within 7 business days from the moment get the goods.
- Please do not use finger nails or hard sharp object to touch the surface of the LCD.
- MYIR suggests user purchasing a piece of special wiper to wipe the LCD after long time use, please avoid clean the surface with fingers or hands to leave fingerprint.

- Do not clean the surface of the screen with chemicals.
- Please read through the product user manual before you using MYIR ' s products.
- For any maintenance service, customers should communicate with MYIR to confirm the issue first. MYIR' s support team will judge the failure to see if the goods need to be returned for repair service, we will issue you RMA number for return maintenance service after confirmation.

Maintenance period and charges

MYIR will test the products within three days after receipt of the returned goods and inform customer the testing result. Then we will arrange shipment within one week for the repaired goods to the customer. For any special failure, we will negotiate with customers to confirm the maintenance period.

For products within warranty period and caused by quality problem, MYIR offers free maintenance service; for products within warranty period but out of free maintenance service scope, MYIR provides maintenance service but shall charge some basic material cost; for products out of warranty period, MYIR provides maintenance service but shall charge some basic material cost and handling fee.

Shipping cost

During the warranty period, the shipping cost which delivered to MYIR should be responsible by user; MYIR will pay for the return shipping cost to users when the product is repaired. If the warranty period is expired, all the shipping cost will be responsible by user.

Products Life Cycle

MYIR will always select mainstream chips for our design, thus to ensure at least ten years continuous supply; if meeting some main chip stopping production, we will inform customers in time and assist customers with products updating and upgrading.

Value-added Services

- MYIR provides services of driver development base on MYIR' s products, like serial port, USB, Ethernet, LCD, etc.
- MYIR provides the services of OS porting, BSP drivers' development, API software development, etc.
- MYIR provides other products supporting services like power adapter, LCD panel, etc.
- ODM/OEM services.

MYIR Electronics Limited

Room 04, 6th Floor, Building No.2, Fada Road,
Yunli Inteiligent Park, Bantian, Longgang District.

Support Email: support@myirtech.com

Sales Email: sales@myirtech.com

Phone: [+86-755-22984836](tel:+86-755-22984836)

Fax: [+86-755-25532724](tel:+86-755-25532724)

Website: www.myirtech.com